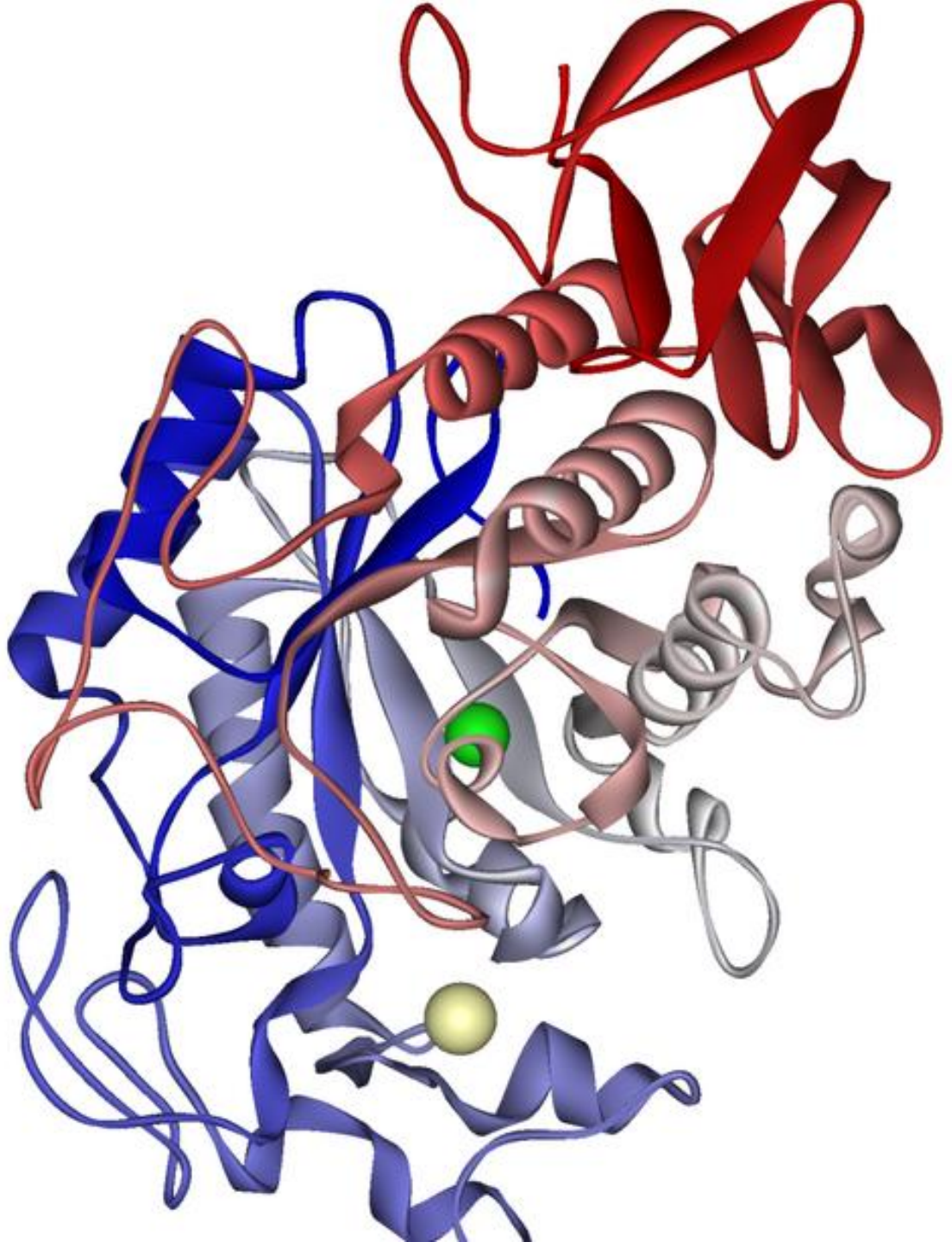
DP Biology – C1.1 (HL extension)

Controlling metabolism

- How do enzymes work with other molecules?
- Which parts of metabolism depend on one another?



HL statements C1.1.11–17. Textbook pp. 396–404.
Image: salivary amylase, an extracellular enzyme. Public domain.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	C1.1.11–13	Compare intracellular and extracellular reactions, explain why metabolism releases heat, and compare linear and cyclical pathways
2	C1.1.14–15	Compare non-competitive (allosteric) and competitive inhibition, using statins, and interpret data
3	C1.1.16–17	Explain feedback inhibition (isoleucine) and mechanism-based inhibition (penicillin)

The big picture: keeping metabolism under control



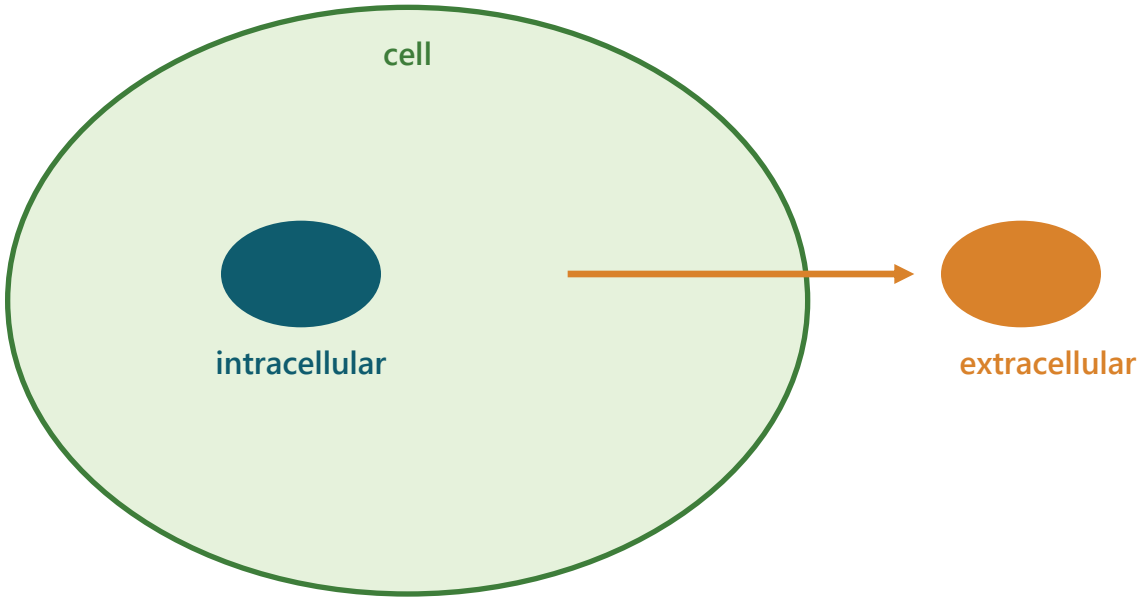
A cell runs thousands of reactions. It controls them by **where** enzymes work and by molecules that **inhibit** enzymes.

Big question

How can one molecule switch off a whole pathway, and how do drugs exploit this?

Where enzymes work

C1.1.11



	Intracellular	Extracellular
Where	Inside the cell	Outside
Example	Glycolysis, Krebs	Gut digestion
Made on	Free ribosomes	Rough ER; secreted

Think

Fungi digest food outside their bodies. Why is secreting enzymes useful for an organism that lives in its food?

Metabolism always releases heat



Emperor penguins. Ian Duffy, CC BY 2.0.

No energy transfer is 100% efficient: some energy is always lost as **heat**. Respiration stores only about a third of glucose's energy in ATP.

Endotherms use this heat to keep warm.



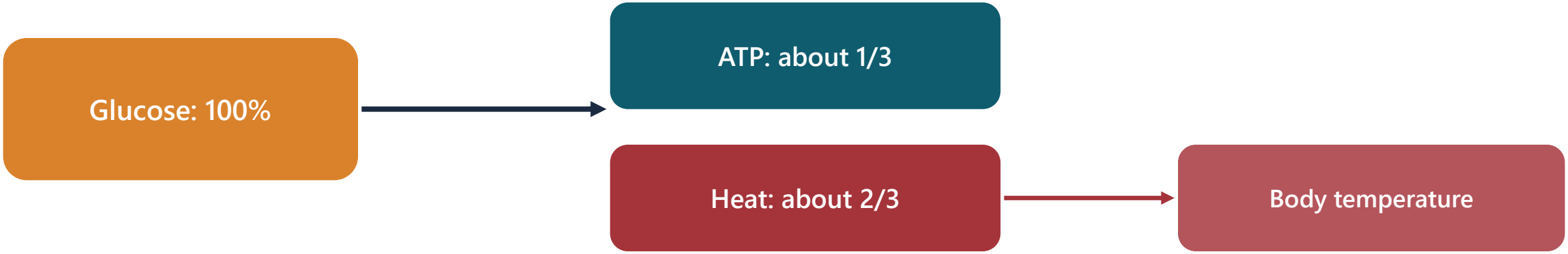
A steaming compost heap. Albarubescens, CC BY-SA 4.0.

Question

Why does a large compost heap get hotter than a small one?

Where the energy goes

C1.1.12



Mammals and birds keep warm with this heat. **Shivering** raises it; **brown fat** respires to release heat directly.

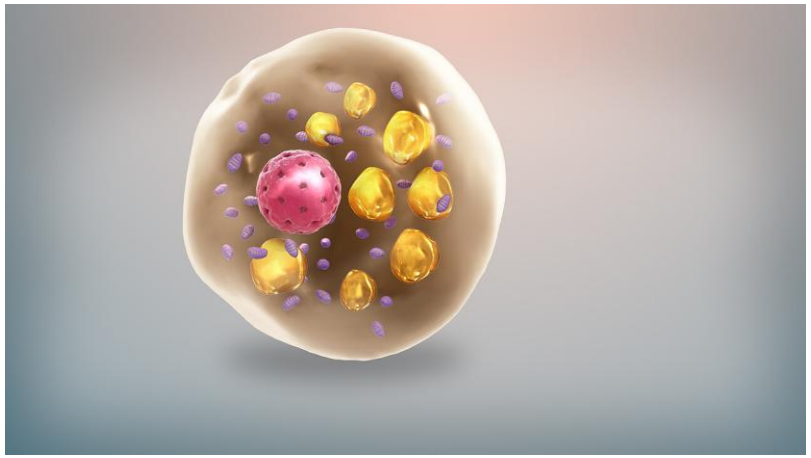
Link
C3.1 / D3.3: thermoregulation.

Brown fat: heat on purpose

C1.1.12

Brown fat cells have many mitochondria. They respire without making much ATP, so nearly all the energy is released as **heat**.

Babies and hibernating mammals rely on it.



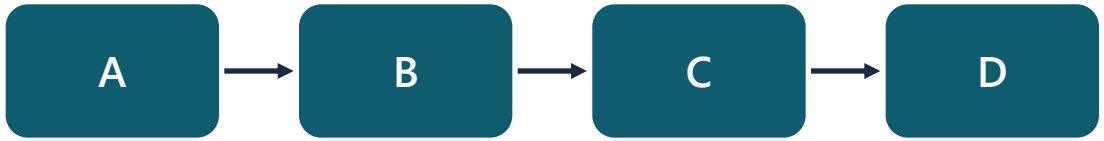
A brown fat cell (illustration). Scientific Animations, CC BY-SA 4.0.

Think

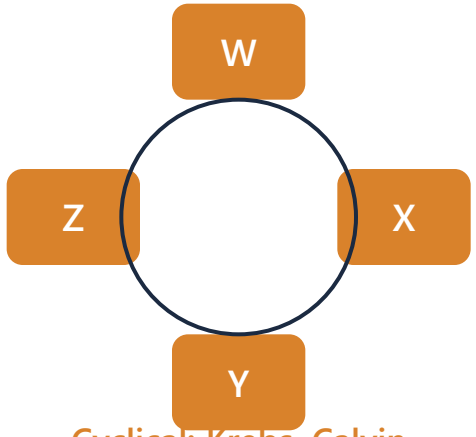
Why does a brown fat cell need so many mitochondria and lipid droplets?

Linear and cyclical pathways

C1.1.13



Linear: glycolysis



Cyclical: Krebs, Calvin

	Linear	Cyclical
Shape	Substrate to end product	Last step regenerates the first
Examples	Glycolysis	Krebs cycle; Calvin cycle

Key idea

Each step has its own enzyme, so each step can be controlled.

Practice: where, heat and pathways

C1.1.11–13

Classify

Amylase in saliva; ATP synthase: intra or extra?

Explain

Why active muscles warm the body.

Classify

Glycolysis and the Krebs cycle: linear or cyclic?

Use

intracellular, extracellular, heat, cyclical

Check-in: class 1

C1.1.11–13

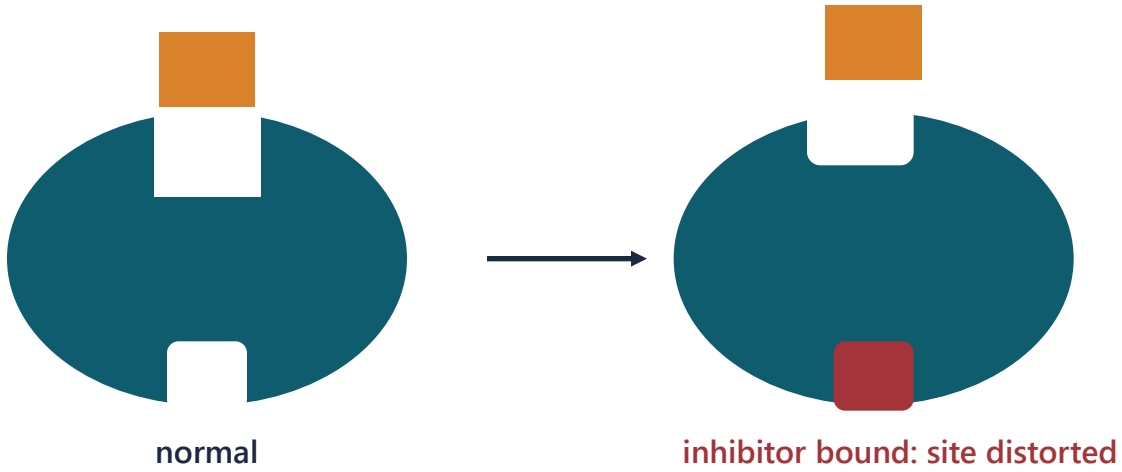
1. Give an example of an intracellular and an extracellular reaction.
2. State where secreted enzymes are made.
3. Explain why metabolism releases heat.
4. Distinguish linear and cyclical pathways.
5. Explain why each pathway step can be controlled.

How to use this

Answer without your notes, then check against slides 3–9.

Non-competitive inhibition at an allosteric site

C1.1.14



An **allosteric site** is away from the active site. An inhibitor binding there **changes the active site's shape**.

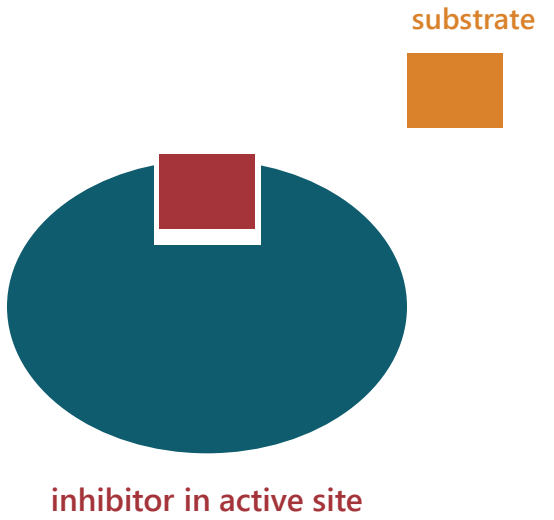
Usually reversible. **More substrate does not help**: the maximum rate falls.

Example

Heavy-metal ions such as mercury and lead.

Competitive inhibition

C1.1.15



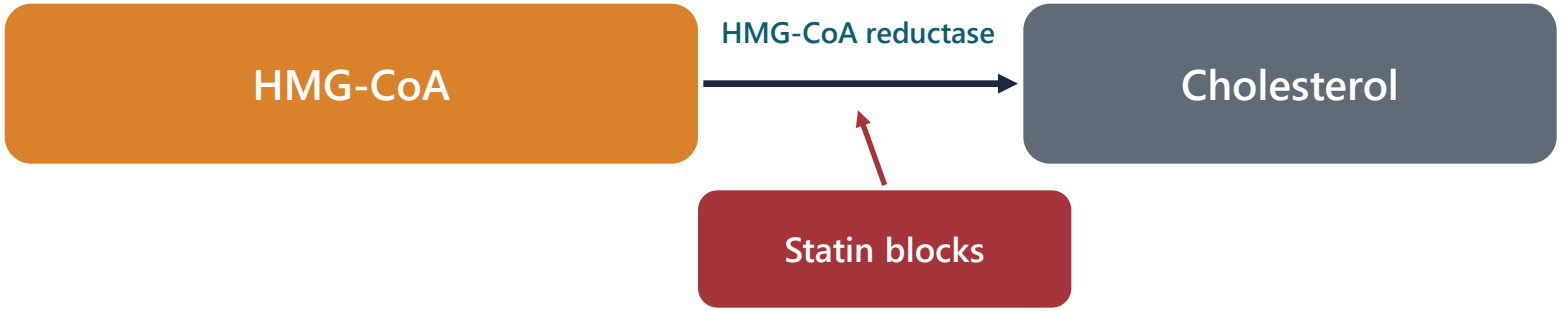
A **competitive inhibitor** resembles the substrate and binds the **active site**, blocking it.

More substrate makes substrate binding more likely, so the maximum rate can still be reached.

	Competitive	Non-competitive
Binds to	Active site	Allosteric site
More substrate	Overcomes it	No effect

Competitive inhibition: statins

C1.1.15



Statins resemble HMG-CoA and compete for the active site of **HMG-CoA reductase** in the liver.

Less cholesterol is made, so blood LDL falls, so heart attacks become less likely.

In medicine

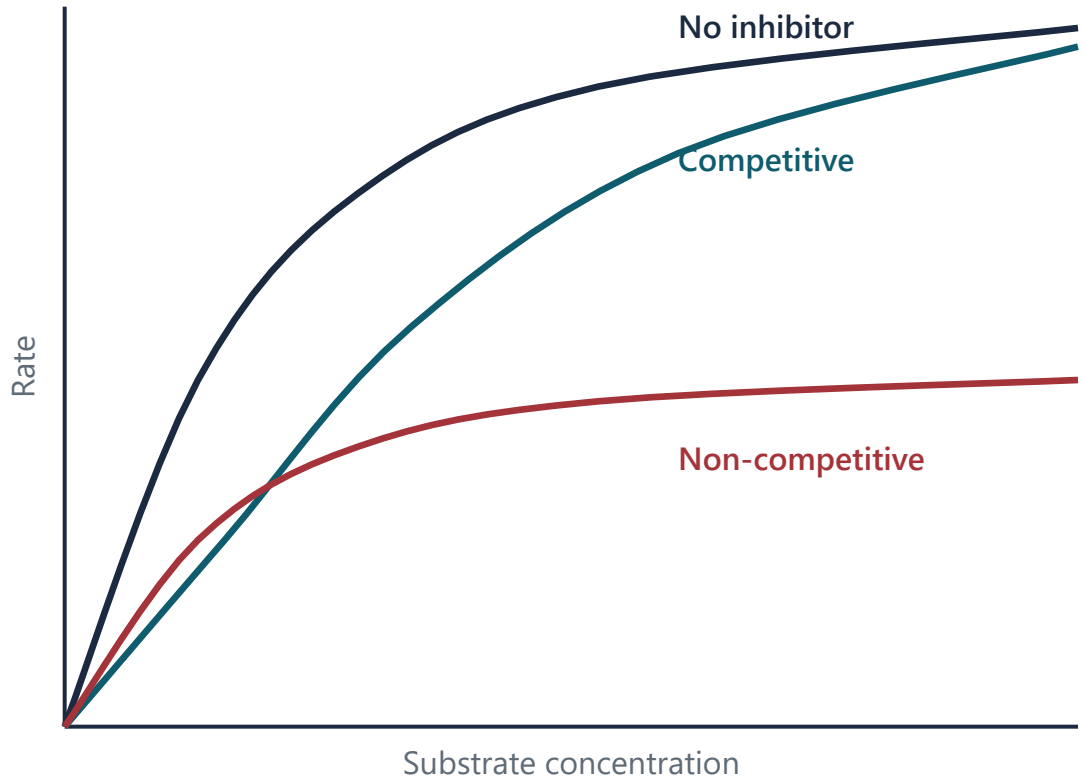
Statins are among the most prescribed drugs in the world.

Key idea

A drug shaped like the substrate can switch off one step of a pathway.

Telling inhibitors apart from data

C1.1.14–15



Substrate	None	X	Y
2	40	15	20
10	88	62	44
40	100	97	50

Data

Invented rates (% of uninhibited maximum). Which inhibitor, X or Y, is competitive? Use the high-substrate data.

Comparing the two

C1.1.14–15

Competitive	Non-competitive	Both
Active site; similar shape; overcome by substrate	Allosteric site; changes shape; max rate lowered	Usually reversible: inhibitor leaves again

Sketch

Draw rate against substrate for no inhibitor, competitive and non-competitive, and label the maximum rates.

Practice: inhibition

C1.1.14–15

Predict

Double the substrate with a statin present.

Explain

Why lead lowers the maximum rate.

Design

How to test if a new drug is competitive.

Use

active site, allosteric, maximum rate, compete

Check-in: class 2

C1.1.14–15

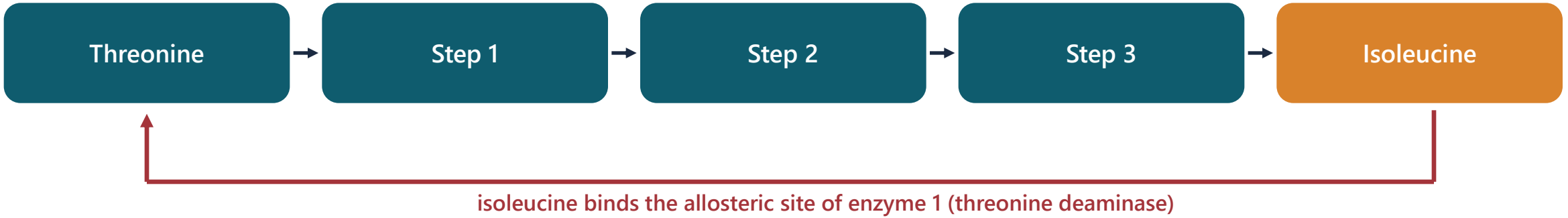
1. Define allosteric site.
2. Distinguish competitive and non-competitive inhibition.
3. Explain how statins lower cholesterol.
4. Explain why more substrate overcomes competitive inhibition.
5. Identify the inhibitor type from a graph that plateaus lower.

How to use this

Answer without your notes, then check against slides 11–16.

Feedback inhibition: isoleucine

C1.1.16



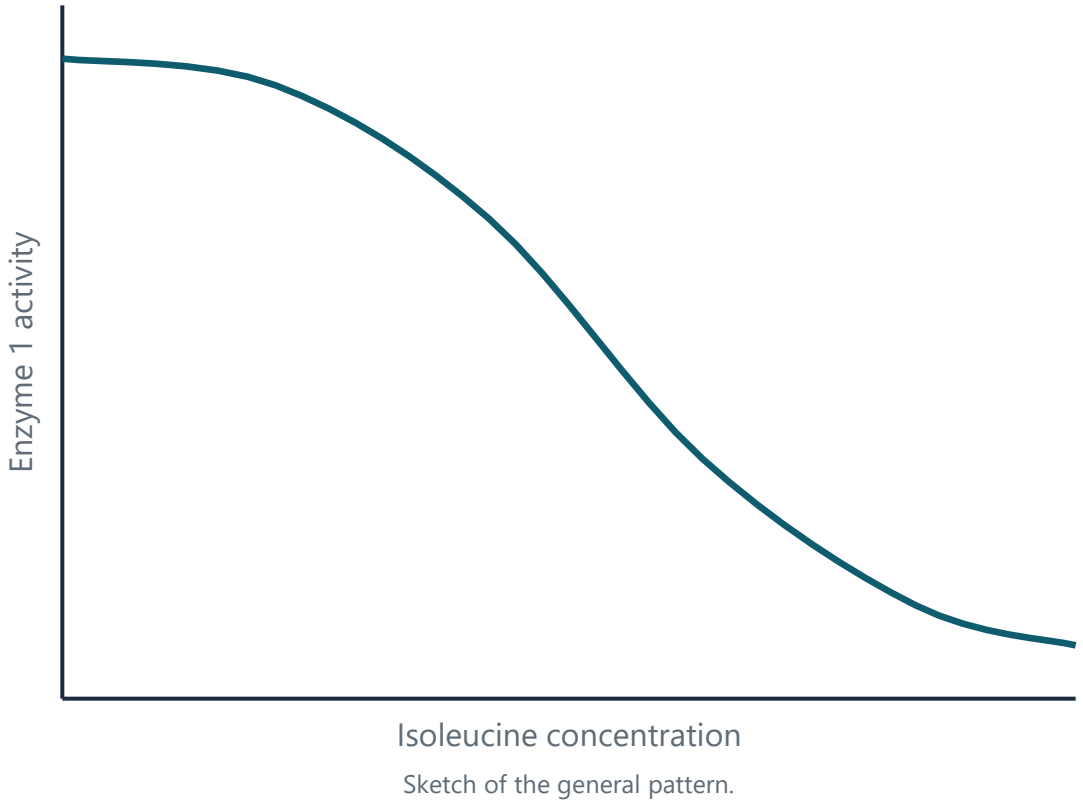
Excess isoleucine **switches off** its own pathway. When it is used up, it leaves and the pathway restarts: **negative feedback**.

Why the first enzyme?

Stopping step 1 stops all intermediates building up; no energy or threonine is wasted.

Feedback in action

C1.1.16



Sketch of the general pattern.

The more isoleucine, the **lower** the activity of enzyme 1.

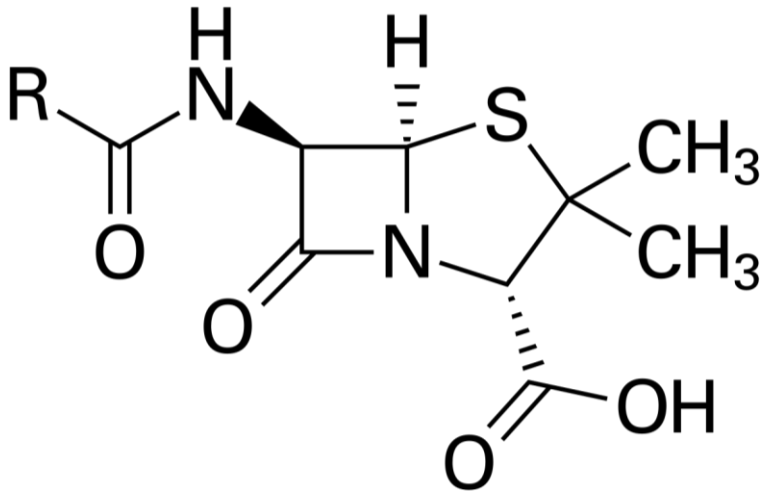
Supply matches demand automatically.

Try it

Predict the effect on the pathway if a cell starts making lots of protein.

Mechanism-based inhibition: penicillin

C1.1.17



Core structure of penicillin. Public domain.

Bacterial **transpeptidases** cross-link cell-wall peptidoglycan.

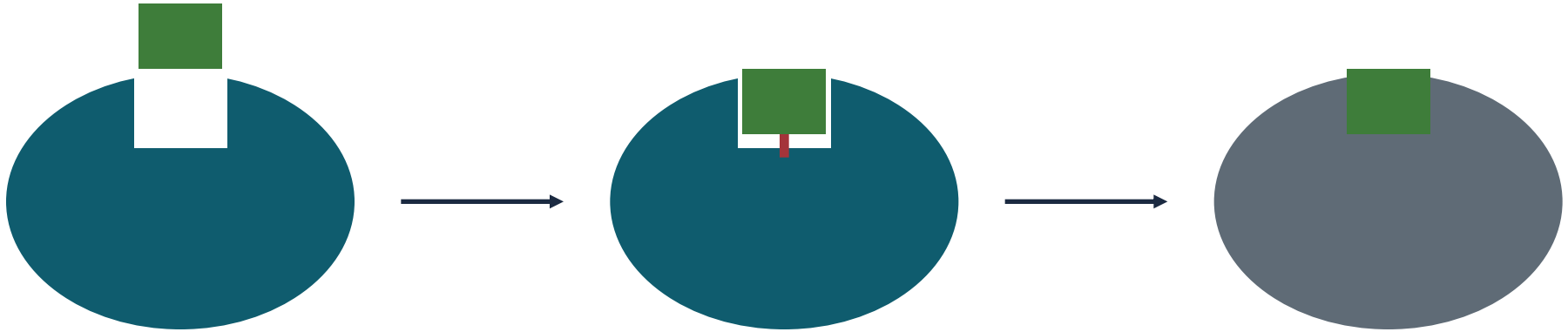
Penicillin enters the active site; its strained ring opens and forms a **covalent bond**, **permanently** inactivating the enzyme. Growing bacteria burst.

Key idea

Irreversible: the enzyme takes part in its own inactivation.

What happens in the active site

C1.1.17



1. penicillin enters

2. covalent bond forms

3. enzyme dead for good

Compare

Competitive inhibitors leave the active site again. Penicillin never leaves, so the bacterium must make new enzyme.

How bacteria resist penicillin

C1.1.17

Mechanism	How it works
Changed transpeptidase	A different active site (e.g. MRSA): penicillin binds poorly
Penicillinase	Breaks penicillin’s ring before it acts

Evolution

Resistant bacteria survive and reproduce, so resistance spreads (D4.1). Overuse speeds this. Clavulanic acid blocks penicillinase in some drugs.

Four ways to slow an enzyme

C1.1.14–17

Type	Binding site	Reversible?	Example
Competitive	Active site	Yes	Statins
Non-competitive	Allosteric site	Yes	Heavy-metal ions
Feedback	Allosteric site	Yes	Isoleucine
Mechanism-based	Active site, covalent	No	Penicillin

Think

Why must a cell replace an enzyme hit by penicillin, but not one hit by a statin?

Check-in: class 3

C1.1.16–17

1. Explain feedback inhibition using isoleucine.
2. Explain why the first enzyme is inhibited.
3. Explain how penicillin inactivates transpeptidase.
4. Give two ways bacteria resist penicillin.
5. Explain why mechanism-based inhibition is irreversible.

How to use this

Answer without your notes, then check against slides 18–23.

Common mistakes to avoid

Not this	But this
Non-competitive inhibitors block the active site.	They bind an allosteric site.
More substrate overcomes all inhibitors.	Only competitive inhibition.
Feedback acts on the last enzyme.	Usually on the first.
Penicillin kills human cells.	Human cells have no walls.
All glucose energy becomes ATP.	About a third; the rest is heat.

Summary: controlling metabolism

C1.1.11–17

	Key idea
C1.1.11	Intracellular vs extracellular
C1.1.12	Heat lost; endotherms use it
C1.1.13	Linear vs cyclical pathways
C1.1.14	Allosteric: non-competitive
C1.1.15	Competitive: statins
C1.1.16	Feedback: isoleucine
C1.1.17	Mechanism-based: penicillin

Check that you can...

1. **Compare** intracellular and extracellular reactions
2. **Explain** why metabolism releases heat
3. **Distinguish** linear and cyclical pathways
4. **Compare** competitive and non-competitive inhibition
5. **Explain** feedback inhibition
6. **Explain** mechanism-based inhibition and resistance

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Give one example each of an intracellular and an extracellular enzyme reaction.
2. Explain why increasing substrate concentration reduces the effect of a statin.
3. State why penicillin is called a mechanism-based inhibitor.

Image credits

Salivary alpha-amylase 1SMD.png – Own work. – Public domain

Aptenodytes forsteri -Snow Hill Island, Antarctica -adults and juvenile-8.jpg – Ian Duffy from UK – CC BY 2.0

Steam from compost heaps.jpg – Albarubescens – CC BY-SA 4.0

Brown fat cell.jpg – https://www.scientificanimations.com/ – CC BY-SA 4.0

Penicillin core.svg – Yikrazuul – Public domain

All from Wikimedia Commons (commons.wikimedia.org, search the file name). Some images are cropped. CC licences: creativecommons.org/licenses